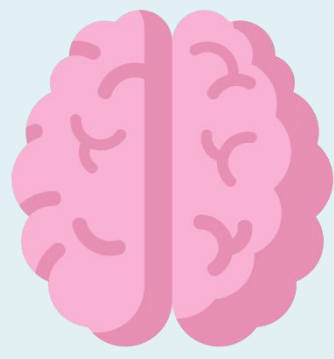
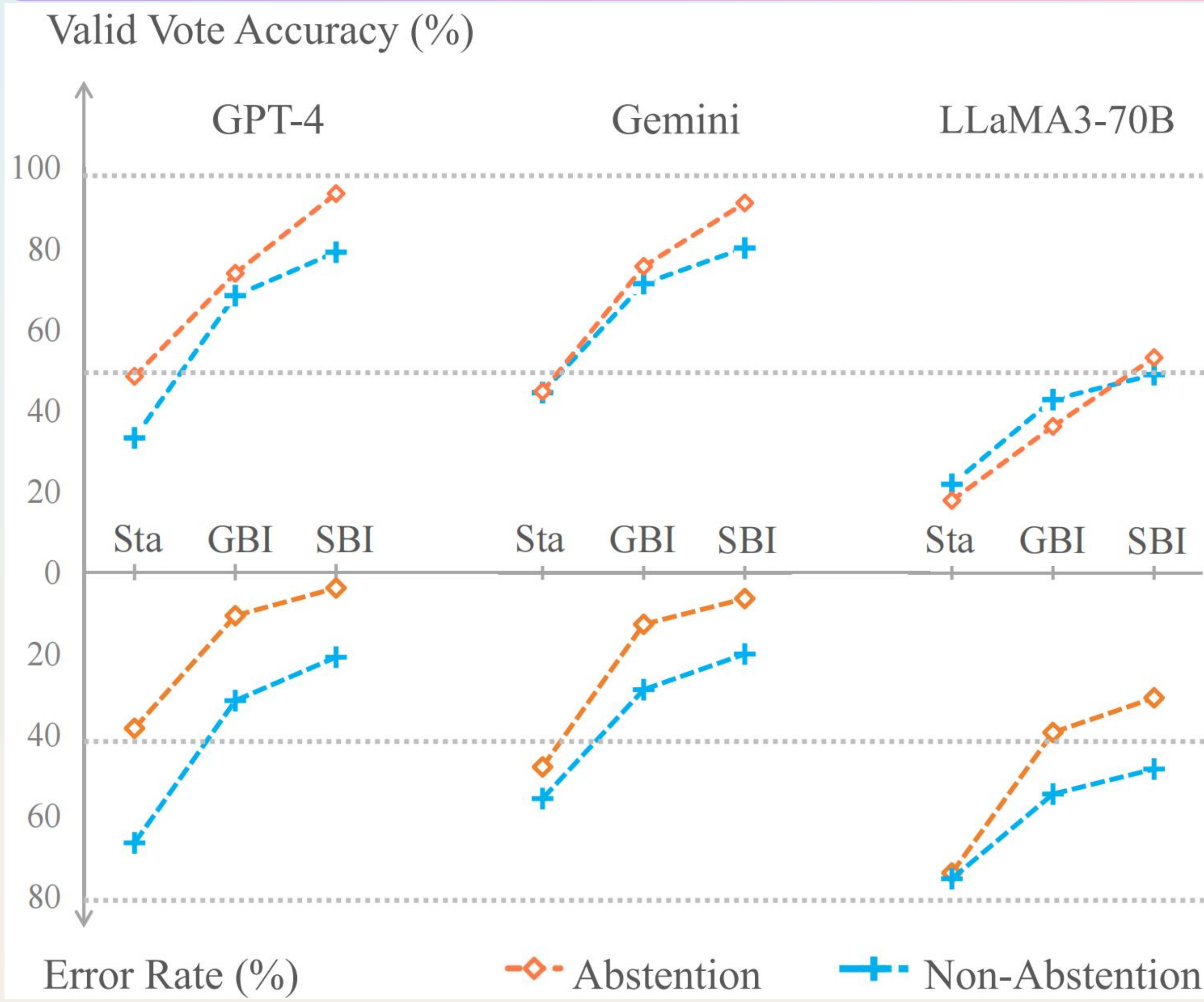


Balancing Rigor and Utility: Mitigating Cognitive Biases in Large Language Models for Multiple-Choice Questions

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INTRODUCTION



Bias-aware prompting enables safer, more human-aligned LLMs.

Valid Vote Accuracy & Error Rate across Models and Prompting Strategies
SBI (Specific Bias Inspection) combined with abstention achieves the highest accuracy and lowest error rate across all three LLMs. The results demonstrate that specific bias awareness enhances decision precision while strategic abstention prevents error-prone responses.

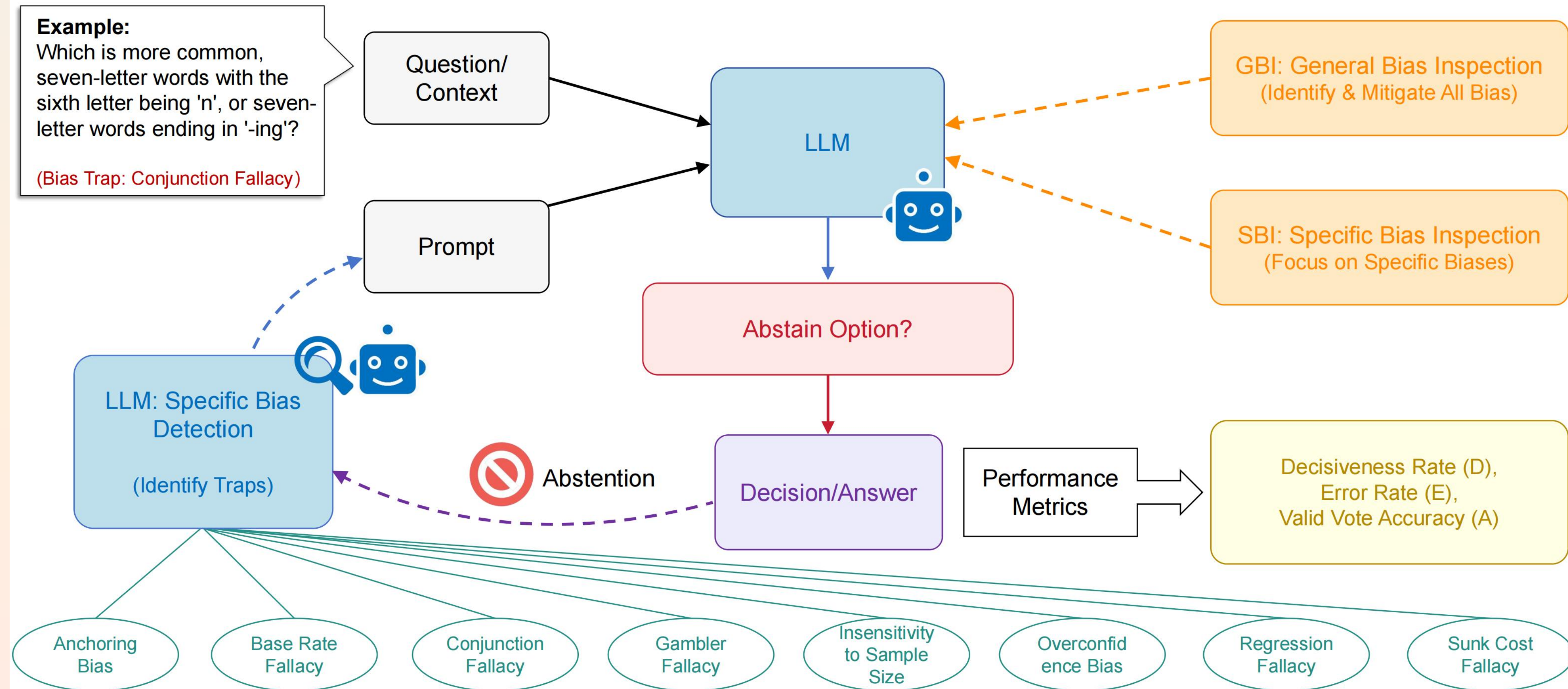
Bias-Aware Feedback Loop
Upon uncertainty, bias cues from the DetectModel guide the AnswerModel toward more confident and reliable decisions.

METHODOLOGY

Algorithm 1 Feedback Loops with Bias Detection

```
1: Initialize: Bias  $\leftarrow$  None, LoopCount  $\leftarrow$  0
2: Decision  $\leftarrow$  ANSWERMODEL(MCQ)
3: while LoopCount < MaxLoops and Decision = Abstain do
4:   Bias  $\leftarrow$  DETECTMODEL(MCQ)
5:   Decision  $\leftarrow$  ANSWERMODEL(MCQ, Bias)
6:   LoopCount  $\leftarrow$  LoopCount + 1
7: end while
8: Output: Decision
```

LLM Decision Pipeline with Bias Detection & Abstention

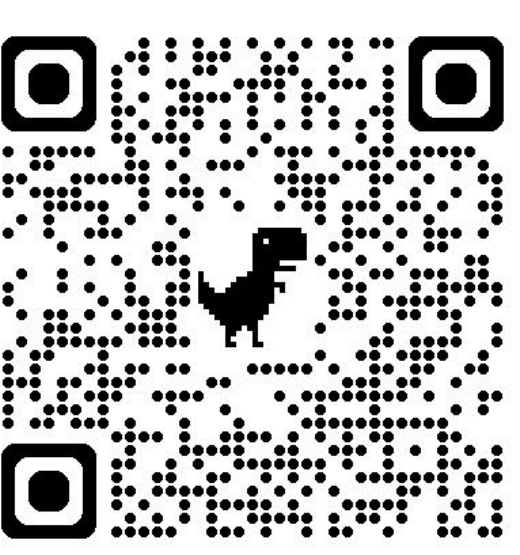


The framework introduces a dynamic decision flow, combining general/specific bias inspection, abstention, and a bias detection module. This allows the LLM to identify traps and abstain when necessary, improving reasoning under uncertainty.

Reasoning Outcomes under Different Prompting Strategies
Prompting strategies significantly impact reasoning outcomes. SBI prompts increase TT (true reasoning with correct answer) while reducing FF (false reasoning with incorrect answer). Abstention also helps avoid low-confidence errors.

Abstention Rate Trends across Bias Types
SBI reduces over-cautious abstention compared to GBI by focusing on specific bias types. This suggests targeted prompting encourages confident decisions while still mitigating known traps.

EXPERIMENT RESULT



Scan to access the BRU dataset, code, and full paper!